CHENFENG LI

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Statistics graduate with experience in questionnaire design, statistical consulting, and applied research using Python and R. Completed SAS Programmer and Advanced Programmer Professional Certificates covering DATA step programming, PROC SQL, macros, and data preparation.

EDUCATION

MS Statistics | University of Chicago Sep 2022 - Jun 2024

GPA: 3.73/4.00. Tuition Scholarship of the Statistics Master Program, 2022 and 2023.

BS Mathematics | The Chinese University of Hong Kong Sep 2018 - Jul 2022

First Class Honours. Concentration: Computational Big Data Analytics. Minor: Statistics.

ACADEMIC RESEARCH AND CONSULTING

Media Images and Political Bias | Master's Thesis, UChicago Sep 2023 - May 2024

Examined whether U.S. news outlets with different political orientations portray politicians with different facial expressions. Combined automated image analysis with statistical comparisons to investigate patterns in media image selection. Prof. Aaron Schein was the advisor of this master's thesis.

- Processed 1,019,845 news-image urls and retained 842,232 images after cleaning and source filtering, covering eight U.S. news outlets across comprehensive political bias.
- Developed a politician name recognition model via collecting labelled images from Wikimedia, embedding with FaceNet, and identifying with nearest-neighbor method. Obtained 80.6% accuracy in 9308 politicians.
- Acquired the politician's names from raw image with this model and extracted seven-dimensional emotion features from a pretrained expression classification model RMN.
- Compared expression patterns for each politician in different media using UMAP, PCA, and classification models with validation sets and confusion matrices. Found limited separation by political orientation.

Statistical Consulting for Clinical and Health Services Studies | UChicago Sep - Dec 2022

Worked in a five-person consulting team on three client studies; clarified research questions in client meetings, reviewed data and contributed actionable statistical recommendations.

- Advised on logistic regression, covariate selection, and patient grouping for a study of medication and ventilation outcomes involving approximately 5,000 hospitalized COVID-19 patients.
- Reviewed modeling options for a home and community-based services study with approximately 700,000 observations from 20 hospitals, distinguishing association from causal objectives when considering fixed effects and propensity-score weighting.
- Helped plan a pediatric CPR study by recommending regression models and sample collection targets, identifying redundant covariates, and explaining analysis choices to the clinical collaborator.

Public Perceptions of Chinese Medicine | Capstone Study, CUHK Jun - Dec 2021

Examined the current state and future development of Chinese medicine from healthcare, economic, and cultural perspectives in a four-person team, combining literature review, expert interviews, and public surveys.

- Led the survey and quantitative-analysis component; designed a questionnaire and coordinated online and paper distribution with teammates, collecting 153 responses.
- Analyzed and visualized survey responses in Python to examine public attitudes toward Chinese medicine and differences by age and education, informing the study's discussion of current situation.
- Helped design interview questions for Chinese medicine professors and practitioners to explore professional perspectives on the field's current challenges and future development.
- Contributed survey findings to the group report and presentation, discussing nonresponse, uneven demographic representation, and omitted variables that limited interpretation and generalization.

SELECTED ACADEMIC PROJECTS

Modified Attention Kernels and Few Shot Learning | UChicago Oct - Dec 2023

Investigated whether alternative attention mechanisms could improve GPT-2 on few-shot learning in a three-person team. Contributed to model modification and evaluated performance on science reasoning benchmarks.

- Modified attention in nanoGPT models from standard dot-product kernel to Gaussian, polynomial, and periodic kernels. Trained the three models and a baseline model on OpenWebText to reproduce GPT-2.
- Prepared the AI2 Reasoning Challenge dataset and implemented evaluation of the baseline and three modified models on science reasoning questions.
- Compared ARC results alongside the team's other benchmarks; standard dot-product attention performed best overall, while nonlinear kernels varied across tasks.
- Presented a project poster at the course seminar and discussed the findings with students and faculty.

Robustness to Spurious Correlations | UChicago Apr - May 2023

Led a three-person study investigating whether distributionally robust optimization (DRO) could reduce neural networks' reliance on misleading correlations and improve predictions when those patterns no longer hold.

- Constructed a colored-MNIST dataset where digit classes were associated with different colors, creating a misleading shortcut that allowed models to predict from color rather than digit shape.
- Implemented and compared standard training through empirical risk minimization (ERM) with group DRO, which prioritizes the worst-performing data group; tested both approaches with and without regularization.
- Found training with ERM failed on underrepresented color–label combinations while DRO achieved high accuracy in all groups. Adding regularization to DRO improved worst-group test accuracy from 44% to 68%.
- Extended the evaluation to image colors absent from training. Regularized DRO maintained similar performance, providing evidence that it relied less on the color shortcut and generalized to unfamiliar examples.

APPLIED RESEARCH AND DATA ANALYSIS PROJECTS

Polli Aerial Image Analysis and Thistle Detection | Packet Digital Mar 2026 - Present

Supported Botlink, Packet Digital's drone software department, in improving Polli's detection of agricultural thistles from aerial imagery through expert-reviewed training data and model refinement.

- Identified that training imagery came from a single location and proposed collecting images across multiple fields to evaluate model generalization.
- Deployed Label Studio on AWS with controlled access, enabling external biologists to review images and annotations while limiting exposure of company data.
- Automated conversion of predictions from the existing model into Label Studio's annotation format. Imported these preliminary labels with high-resolution images into Label Studio for expert review and modification.
- Divided high-resolution images into overlapping patches to preserve small weeds and objects near image boundaries; applied image transformations to expand training-data diversity.
- Fine-tuned the existing YOLO model and compared alternative detectors. Selected the fine-tuned model achieving 90% mAP@0.5, balancing detection performance and compatibility with the existing software.

Early Battery Degradation Prediction and Feasibility Analysis | Packet Digital Oct 2025 – Jan 2026

Supported battery R&D by assessing early lifespan prediction and whether models trained on laboratory datasets could transfer to company battery systems.

- Analyzed approximately 1,500 cycles over 10 months in Python to assess early battery degradation.
- Visualized voltage, current, capacity, and resistance data; used cycle-consistency checks and plots to identify irregular charge-discharge cycles.
- Cleaned cycle data and extracted statistical and temporal features describing degradation.
- Trained an XGBoost baseline on the MATR dataset to predict life to 80% capacity from the first 100 cycles.
- Identified limitations of single-pack data and mismatch with benchmark cells; documented constraints on model transfer and the need for broader data collection.

RESEARCH AND TECHNICAL SKILLS

SAS: DATA step data preparation and merging, PROC SQL, macros, summary reporting, and debugging.

Python and R: Data cleaning, statistical modeling, and visualization; pandas, NumPy, statsmodels, scikit-learn, PyTorch, and R.

Research methods: Questionnaire and interview design and implementation, linear and logistic regression, model validation, subgroup analysis, and interpretation of sampling and measurement limitations.

Data and reporting: SQL, Power BI, Excel, Git, LaTeX, and R Markdown.

Languages: English (fluent); Mandarin Chinese and Cantonese (native).

PROFESSIONAL EXPERIENCE

Maintenance Technician | Packet Digital, Fargo ND — Oct 2025 - Present

- Support Botlink, Packet Digital's drone software department, on Polli aerial-image analysis, including annotation workflows, data preparation, and detection-model evaluation.
- Supported battery R&D development through lifecycle-data analysis and an early battery-life prediction study.
- Coordinated with contractors and engineers on cell plant construction.
- Performed routine and timely equipment operation and maintenance.

Data Engineer | Beaconfire, West Windsor NJ — Feb 2025 – Aug 2025

- Built and automated an AWS-based clinical workload monitoring pipeline using PySpark, SQL, and Snowflake, and developed Power BI dashboards to track provider workloads and staffing efficiency, enabling better resource allocation and reducing reporting time by over 80%.

Data Scientist | SynergisticIT, Fremont CA, remote — Jun 2023 – Mar 2025

- Built an Azure-based sentiment analysis pipeline using PySpark, Databricks, and BERT to analyze 16,000 customer reviews, automating sentiment classification and reporting to support product and marketing decisions.
- Developed a sales forecasting model using Python, XGBoost, and LightGBM to analyze 2.9 million sales records, predict future sales, and support inventory optimization across 59 stores.
- Developed an interactive Power BI sales dashboard using SQL and DAX to track sales performance, discounts, and regional trends, enabling data-driven business decisions.
- Built a Power BI retail transaction dashboard using Power Query and DAX to analyze product sales, customer behavior, and seasonal trends, providing actionable insights for retail planning and promotions.

SELECTED CERTIFICATES AND TRAINING

SAS Advanced Programmer Professional Certificate - SAS via Coursera | Dec 2024

SAS Programmer Professional Certificate - SAS via Coursera | Oct 2024

AWS Certified Cloud Practitioner – AWS | May 2024

Power BI Data Analyst Associate - Microsoft | Mar 2024

Microsoft Office Specialist: Excel Associate (Office 2019) - - Microsoft | Mar 2024

Google Advanced Data Analytics Certificate - Google via Coursera | Jun 2023